Lalitha Balachandran

last updated September 2026

: Department of Ecology & Evolutionary Biology
University of California, Santa Cruz
130 McAllister Way, Santa Cruz, CA 95060

✉: lalithab@ucsc.edu
: lalitha-balachandran.github.io

Interests

Pedagogy: instructional design, inquiry-based & experiential learning
Research: psycholinguistics, language comprehension, memory & metacognition, instructional equity, institutional research

Education

2019-24	**Ph.D. Linguistics**, University of California, Santa Cruz *Dissertation*: Memory knows its bounds: Encoding contexts in sentence comprehension *Advisor*: Dr. Matt Wagers
2016-18	**B.A. Linguistics, Specialization in Computing**, University of California, Los Angeles
2015-16	Diablo Valley College, Las Positas College
2014	University of Colorado, Boulder

Academic Employment

2024-26	**Postdoctoral Scholar**, "FUTURE" Education Research Team Department of Ecology & Evolutionary Biology, University of California, Santa Cruz *Grant:* Field-based Undergraduate Training: Utilizing Research for Equity (FUTURE) in Biology (NSF #2329523) *PIs*: Drs. Roxanne Beltran, Robin Dunkin, & Erika Zavaleta
2018-19	**Lab Manager**, Cognitive Neuroscience of Language Lab Department of Linguistics, University of Maryland, College Park *PI*: Dr. Colin Phillips

Publications & Presentations

Note: † (co-)first author; "in prep" manuscripts are complete drafts being prepared for submission

Publications

in prep	**Balachandran, L.**, Kaplanis, N., Bhatti, H.A., Kouba, P.V., Zavaleta, E., Beltran, R., & Arcila Hernández, L. Describing & designing axes of experiential learning: A design framework for undergraduate field and research experiences.
in prep	Kaplanis, N.J., Arcila Hernández, L.M., Kouba, P., **Balachandran, L.**, Bhatti, H.A., Pfau, M.V., Dayton, G., Croll, D., Gaensler, B., Beltran, R.S., & Zavaleta, E.S. A large-scale comparison of diverse undergraduate experiential programs reveals high-impact practices, variation in student outcomes, and opportunities for curricular complementarity.
submitted	Kouba, P.V., Arcila Hernández, L., Kaplanis, N. J., **Balachandran, L.**, Bhatti, H. A., Beltran, R. S., & Zavaleta, E. Social network strength amplifies benefits of undergraduate ecology field programs.
under review	**Balachandran, L.**, Arcila Hernández, L., Borker, A., Bhatti, H.A., Kouba, P.V., Zavaleta, E., Beltran, R., & Dunkin, R. Experiential learning supports knowledge retention and metacognitive skills transfer in related undergraduate ecology courses.
under review	**Balachandran, L.**, McNamara, M. B., Lee, H. K., & Tassio, M. Impacts of online instruction in prerequisite courses on student performance in postrequisite courses.
in revision	Pfau, M. J., Bhatti, H. A., Arcila Hernández, L. M., Kaplanis, N. J., Borker, A. L., **Balachandran, L.**, Zavaleta, E. S., Dayton, G. H., & Beltran, R. S. Pathways and predictors of independent research at a higher education institution.
2026	Bhatti, H. A., Arcila Hernández, L., **Balachandran, L.**, Kouba, P., Croll, D., Dayton, G., Marnocha, E., Beltran, R. S., & Zavaleta, E. S. Differential Item Functioning Analysis Reveals Equitable Confidence Development in Undergraduate Field Courses.

2026 **Balachandran, L.**, †Bhatti, H. A., Kouba, P., Arcila Hernández, L., Borker, A., Scalice, D., Lepore, T., Esparza, D., Reilly-Sanchez, L., Smith, M., Ignace, D., Hart, C., Holley, N., White, H., Jolley, A., Cho, H., O'Connell, K., Shaulskiy, S., Zavaleta, E., & Beltran, R. S. Fostering connection to people, place, and self through field biology learning experiences.

2026 Arcila Hernández, L.M., Payne, A.R., Kaplanis, N.J., Borker, A.L., Dunkin, R.R., Alissa, L.M., **Balachandran, L.**, Bhatti, H.A., Carr, M., Croll, D., Dayton, G., Flores, R., Gilbert, G., Kouba, P., Kroeker, K., Lay, C., Leonard, H., Lyon, B., Martinez, A., Mehta, R., Miller, T., Parker, I., Pfau, M., Zavaleta, E.S., & Beltran, R.S. So you want to teach a field-based course: A pratical toolkit for teaching teams. In *Integrative Organismal Biology.*

2025 **Balachandran, L.**, Rich, S., & Wagers, M. How linguistic boundaries form encoding contexts in memory: Evidence from temporal order effects. *Proceedings of Cognitive Science 49.*

2024 Zellner, N. E. B., Boyd, P., Villanueva, S., Alexander, V., Scalice, D., **Balachandran, L.**, Nuñez, J., Hart, B., Gonzales, E., & Ferrell, T. NASA'S SMD Bridge Program: A co-created program that funds faculty and students at under-resourced emerging research institutions. In *55th Lunar and Planetary Science Conference* (Vol. 3040, p. 2550).

2022 Duff, J. & **Balachandran, L.** Participants seek shared outlooks in non-canonical disagreements: Evidence from a corpus of dyadic conversation in English. In the Proceedings of *SemDial* 26 (DubDial).

2022 **Balachandran, L.** Reciprocal questions and the pragmatics of argument reversing verb phrase ellipsis. In the Proceedings of *Sinn und Bedeutung* 26.

Edited Volumes

2022 Balachandran, L. & Duff, J. (eds). Syntax and Semantics at Santa Cruz, Volume V. Linguistics Research Center, UC Santa Cruz.

Other Manuscripts

2024 Smith, M., Gamble, R., Jorgenson, R., Moore Jr, C., Pando, J., Boyd, P., & Zellner, N. (eds.) with **Balachandran, L.** and co-authors. Report from the Bridge Program Workshop Organizing Committee. NASA Science Mission Directorate. [link]

Talks

2026 *Integrating science education research with structural changes in higher education: a case study in field-based learning.* with Lina Arcila Hernández, Erika Zavaleta, Haider Ali Bhatti, Paige Kouba, Abraham Borker, Nikolas Kaplanis, Madison Pfau, Robin Dunkin, & Roxanne Beltran. Talk at the Ecological Society of America Conference.

2026 †*Impacts of field-based and research-based experiential learning on metacognitive skills transfer in undergraduate ecology students.* with Lina Arcila Hernández, Abraham Borker, Haider Ali Bhatti, Paige Kouba, Roxanne Beltran, Erika Zavaleta, & Robin Dunkin. Talk at the Society for the Advancement of Biology Education Research Conference.

2025 †*Metacognitive skills transfer from field-based learning to lecture-based courses.* with Haider Ali Bhatti, Paige Kouba, Lina Arcila Hernández, Roxanne Beltran, Erika Zavaleta, & Robin Dunkin. Talk at the Ecological Society of America Conference.

2025 *Better measuring the impact of field-based learning in undergraduate ecology through Differential Item Functioning (DIF) analysis.* with Haider Ali Bhatti, Lina Arcila Hernández, Paige Kouba, Donald Croll, Gage Dayton, Erin Marnocha, Roxanne Beltran, & Erika Zavaleta. Talk at the Ecological Society of America Conference.

2025 *Social network analyses demonstrate community-building effects of ecology field programs.* with Paige Kouba, Nikolas Kaplanis, Lina Arcila Hernández, Haider Ali Bhatti, Roxanne Beltran, & Erika Zavaleta. Talk at the Ecological Society of America Conference.

2025 †*How linguistic boundaries form encoding contexts in memory: Evidence from temporal order effects.* with Stephanie Rich & Matt Wagers. Talk at CogSci 49.

2024 *NASA's SMD Bridge: A Co-Created Program in Service to Faculty and Students at Under-Resourced, Emerging Research Institutions.* with Daniella Scalice, Padi Boyd, Steven Villanueva,...Hilarie Davis. Talk at Astrobiology Science Conference 2024.

2024 *NASA'S SMD Bridge Program: A Co-Created Program that Funds Faculty and Students at Under-Resourced Emerging Research Institutions.* with Nicolle Zellner, Patricia Boyd, Steven Villanueva, Vemitra Alexander, Daniella Scalice, Jeremias Nuñez, Bri Hart, Edward Gonzalez, & Trena Farrell. Talk at Lunar & Planetary Sciences Conference 2024.

2023 †*Does memory for focus structure interfere with memory for prosody?* with Morwenna Hoeks, Nicholas Van Handel, & Amanda Rysling. Talk at CAMP5.

2023 †*Implicit prosodic representations do not affect recognition or retrieval.* with Matt Wagers & Amanda Rysling. Short talk at CAMP5.

2022 *Participants seek shared outlooks in non-canonical disagreements: Evidence from a corpus of dyadic conversation in English.* with John Duff. Talk at SemDial 26 (DubDial), Technological University Dublin.

2022 *"It's alive!": Animacy-based structural expectations rapidly update with context.* with Stephanie Rich, John Duff, Matthew Kogan, Maya Wax Cavallaro, Nicholas Van Handel, & Matt Wagers. Short talk at the Conference on Human Sentence Processing 35, UCSC.

2021 †*Reciprocal Questions and the Pragmatics of Argument Reversing Verb Phrase Ellipsis.* Talk at Sinn und Bedeutung 26, University of Cologne, Germany.

2021 *Singular vs. Plural Themselves: Evidence from the Ambiguity Advantage.* with Nicholas Van Handel, Stephanie Rich, & Amanda Rysling. Short talk at the CUNY Sentence Processing Conference 34, University of Pennsylvania.

2019 *Number Attraction in Production: errors and speech rate profiles narrow a production-comprehension contrast.* with Margaret Kandel, Cassidy Wyatt, & Colin Phillips. Talk at the CUNY Sentence Processing Conference 32, University of Colorado, Boulder.

Posters

2025 †*Downstream effects of online learning.* with Megan McNamara, Herbert Lee, & Michael Tassio. UCSC Teaching Symposium.

2024 †*Linguistic boundaries delineate contextual domains in memory.* with Matt Wagers. AMLaP 30, University of Edinburgh.

2024 †*Linguistic boundaries reduce encoding interference in temporal order memory.* with Stephanie Rich & Matt Wagers. AMLaP 30, University of Edinburgh.

2024 †*Memory for violated expectations in semantic focus.* with Morwenna Hoeks, Nicholas Van Handel, Cal Boye-Lynn, & Amanda Rysling. HSP 37, University of Michigan, Ann Arbor.

2024 †*Different structural boundaries have different effects on dependency resolution.* with John Duff, Pranav Anand, & Amanda Rysling. HSP 37, University of Michigan, Ann Arbor.

2023 *Enhancing STEM diversity efforts: Insights from geographic analysis of registrants for NASA's SMD Bridge Program Workshop.* with Jeremias Nuñez, Nicolle Zellner, & Lorenzo Pappas. National Diversity in STEM 2023 Conference (SACNAS).

2023 †*Domain-sensitivity of sentence memory and (lack of) temporal contiguity effects.* with Stephanie Rich & Matt Wagers. Architectures & Mechanisms in Language Processing (AMLaP) 29, Basque Center on Cognition, Brain, & Language.

2023 †*Top-down expectations vs. bottom-up information in prosodic memory.* with Morwenna Hoeks, Nicholas Van Handel, & Amanda Rysling. HSP 36, University of Pittsburgh.

2022 †*Compression vs. partition: Memory domains and the processing of appositives.* with John Duff, Pranav Anand, & Amanda Rysling. AMLaP 28, University of York.

Invited Presentations

2026 *Field-based teaching and learning through the lens of educational impact and pedagogical practice.* Experiential Learning Faculty Fellows Workshop, UCSC Teaching & Learning Center.

2025 *Field-based teaching and learning through the lens of educational impact and pedagogical practice.* UC Santa Cruz Ecology & Evolutionary Biology Department Seminar.

2025 *Cultivating holistic mentoring relationships: Getting the support you need.* Center to Advance Mentored, Inquiry-based Opportunities in Ecology & Conservation, Workshop Series, UC Santa Cruz.

2025 *Field-based Undergraduate Training: Utilizing Research for Equity (FUTURE) in Biology.*

STEM Teaching & Learning Community, UCSC Teaching & Learning Center.

2023 *Working with Teaching Teams.*
Co-panelist, New Faculty Teaching Academy, UCSC.

2023 †*Does memory for focus structure interfere with memory for prosody?* with Morwenna Hoeks.
Linguistics at Santa Cruz, UCSC.

2022 †*The pragmatics of argument-reversing verb phrase ellipsis: a QUD-based account.*
Linguistics at Santa Cruz, UCSC.

2020 *Argument Reversals in Verb Phrase Ellipsis.*
Words with Friends, Stanford University.

Workshops & Organized Sessions

2026 *Enhance your field-based teaching: Implementing a practical teaching toolkit for student success.* with co-authors Haider Ali Bhatti, Lina Arcila Hernández, Erika Zavaleta, & Roxanne Beltran. Workshop presented at SABER West 2026 and Ecological Society of America Conference 2026.

2025 *Inclusive teaching practices for more equitable field-based learning.* with co-organizers Lina Arcila Hernández, Ali Bhatti, Paige Kouba, Erika Zavaleta, Roxanne Beltran and presenters Abraham Borker, Daniella Scalice, Taormina Lepore, David Esparza, Danielle Ignace, & Holly White. Organized Oral Session, Ecological Society of America Conference 2025.

2022 *Report on Asian American and Native American Pacific Islander-Serving Institutions; Native Hawaiian-Serving Institutions; Tribal Colleges and Universities; and Native American-Serving Nontribal Institutions.* NASA Science Mission Directorate Bridge Program Workshop.

Other Professional & Research Experience

2023 **Graduate Student Researcher**, Teaching & Learning Center, UC Santa Cruz
Collaborators: Dr. Herbie Lee, Dr. Megan McNamara, Michael Tassio

2022-23 **Graduate Intern**, MOSAICS Education Program (Director: Dr. Patricia Boyd)
Science Mission Directorate, National Aeronautics & Space Administration (NASA)

2021 **Graduate Researcher**, r/lab (Resumption & Animacy Lab) - Contextual Animacy Group (Co-PIs: Drs. Ivy Sichel, Maziar Toosarvandani, Matt Wagers); NSF BCS #2019804
Department of Linguistics, University of California, Santa Cruz

2017-18 **Undergraduate Research Assistant**, Language Processing Lab (PI: Dr. Jesse Harris)
Department of Linguistics, University of California, Los Angeles

2016-18 **Undergraduate Research Assistant**, White Lab (PI: Dr. Stephanie White)
Dept. of Integrative Biology & Physiology, University of California, Los Angeles

2016-18 **Student Assistant**, Office of Information Technology, UCLA

2015-16 **Office Assistant**, LeadGenius, Berkeley, CA

Teaching & Mentorship

Pedagogical Leadership & Professional Development: Instructional Design

2023 **Summer Graduate Student Instructor Peer Support Team**, Course Design & Delivery Program, Teaching & Learning Center (TLC), UC Santa Cruz

2023 **Course Design & Delivery** - Graduate Certificate Program, TLC, UC Santa Cruz

2023 **Equity-minded Mentoring** - Graduate Certificate Program, TLC, UC Santa Cruz

2022 **Teaching for Equity** - Graduate Certificate Program, TLC, UC Santa Cruz

2022 **Graduate Pedagogy Fellow** - Pedagogical Leadership Certificate, TLC, UC Santa Cruz

Pedagogical Professional Development: Field Teaching in Ecology, Biology, and Environmental Studies

2026 **Advancing Field Safety: Preparation and Leadership** - Workshop, UC Field Safety

2025 **Queer & Trans Inclusive Field Safety** - Workshop, UC Field Safety

2025 **Wilderness Mental Health First Aid** - Workshop, UC Field Safety

2025 **Field-Based Teaching Summit**, Ecology & Evolutionary Biology, UC Santa Cruz

Instructional Designer/Education Specialist, UC Santa Cruz

*(co-)designed novel course/workshop

† designed pedagogical activities and/or conducted educational assessment

Fall 2025	Field Biology in Practice†	Instructor: Abe Borker
Fall 2025	Field-based teacher training*	with FUTURE in Biology team
Summer 2025	Variation in Language Processing*,†	Instructor: Stephanie Rich
Winter 2025	Ecology & Evolution†	Instructor: Baldo Marinovic
Winter 2025	Development & Physiology†	Instructor: Robin Dunkin
Summer 2023	Variation in Language Processing*,†	Instructor: Stephanie Rich

Lead Instructor/Facilitator, UC Santa Cruz

Fall 2025, 2026	Field-based Teacher Training*	Co-facilitator: Lina Arcila Hernández
Fall 2023	Psycholinguistics I	Faculty Mentor: Matt Wagers
Fall 2022 - Winter 2023	Pedagogy of Linguistics*	Faculty Mentor: Ryan Bennett

Teaching Assistant, UC Santa Cruz

Spring 2024	Semantics II	Instructor: Roumyana Pancheva
Winter 2024	Introduction to Linguistics	Instructor: Andrew Angeles
Summer 2023	Variation in Language Processing*	Instructor: Stephanie Rich
Spring 2022	Psycholinguistics I	Instructor: Morwenna Hoeks
Winter 2022	Semantics I	Instructor: Jake Vincent
Fall 2021	Psycholinguistics I	Instructor: Matt Wagers
Summer 2021	Language OnLine	Instructor: Tom Roberts
Spring 2021	Invented Languages	Instructor: Pranav Anand
Winter 2021	Syntax I	Instructors: Jorge Hankamer, Ivy Sichel
Fall 2020	Semantics I	Instructor: Jake Vincent

Mentorship

2022-24	**Mentor**, s/lab, UCSC Linguistics. 2022-23 Undergraduate RAs: Cal Boye-Lynn, Jenna Maas, AJ Testerman 2023-24 Undergraduate RAs: Cal Boye-Lynn, Claire Sakurauchi
2018-19	**Mentor**, Cognitive Neuroscience of Language Lab, UMD Linguistics. Undergraduate RAs: Andrew Merchant, Cassidy Wyatt; co-mentored Macie McKitrick, Katte Luckhurst, Stephanie Pomrenke

Awards & Honors

2026	Ocean Fellowship (Environmental Education), Pangaea Exploration Sailing.
2024	Summer Graduate Dissertation Fellowship, The Humanities Institute, UC Santa Cruz.
2024	Jorge Hankamer Outstanding Graduate Instructor in Linguistics Award, UC Santa Cruz.
2023	Summer GSI Peer Support Team Fellowship, TLC, UC Santa Cruz.
2023	Humanities Division Summer Fellowship, UC Santa Cruz.
2022	Graduate Pedagogy Fellows Fellowship, TLC, UC Santa Cruz.
2021	Humanities Summer Fellowship, UC Santa Cruz.
2020	Summer Regents Fellowship, UC Santa Cruz.
2020	Linguistics Department Fellowship, UC Santa Cruz.
2020	Honorable Mention, Graduate Research Fellowship Program, NSF.
2019-20	Regents Fellowship, UC Santa Cruz.
2017-18	Undergraduate Research Fellows Scholarship, Undergraduate Research Center, UCLA.

Service

2025-26	**Reviewer**, Cognitive Science Proceedings.
2025	**Reviewer**, National Aeronautics and Space Administration.
2025	**Reviewer**, Human Sentence Processing Conference.
2022-24	**Member**, EquiL (Equity in Linguistics), UCSC Ling.
2022-23	**Co-founder/organizer**, EquiL (Equity in Linguistics), UCSC Ling.
2022	**Reviewer**, California Meeting on Psycholinguistics, UCLA.

Skills

Programming languages
proficient in R
familiar with Python, MATLAB, JavaScript, HTML
rusty in C++, Haskell

Languages
English (native)
Tamil (heritage)
German (advanced)

Research methods
eye-tracking, EEG, survey design
institutional data analysis, behavioral research

Software
PCIbex, Praat, ELAN, PsychoPy, LaTeX,
SR Research Experiment Builder, Canvas, Qualtrics